

Smart Greenhouse Adaptive Fuzzy Climate Control System

System description and problem justification

The proposed system is an intelligent greenhouse climate control system that utilizes fuzzy logic controllers to automatically regulate temperature and humidity for optimal plant growth. It adjusts ventilation fan power and misting intensity based on continuous monitoring and incorporates plant growth stages to adapt control behavior. A Mamdani fuzzy controller models expert knowledge, while a Sugeno fuzzy controller is used for performance comparison. The fuzzy inference system operates with input variables of temperature, humidity, and growth stage, yielding output variables of fan power and misting level. An optimization stage employs a Genetic Algorithm to fine-tune membership parameters, aiming to minimize control error and energy consumption. Simulation results confirm the system's effectiveness under various conditions.

Maintaining optimal greenhouse conditions is challenging due to nonlinear dynamics, uncertain weather, and variable plant needs. Traditional PID controllers struggle with these uncertainties, while fixed-threshold systems often lead to inefficiencies. Fuzzy logic, incorporating expert knowledge through linguistic rules, allows systems to manage uncertainty better. The Mamdani controller offers interpretability, and the Sugeno controller enhances computational efficiency. To optimize performance, techniques like Genetic Algorithms can refine membership functions, minimizing control errors and improving energy use. This integrated approach results in a robust greenhouse climate control system suitable for varying environmental conditions.

Selected Plant Species

1. **Tomato** – a warm-season crop, sensitive to temperature changes
2. **Lettuce** – a cool-season leafy vegetable, sensitive to heat
3. **Orchid** – a high-humidity flowering plant with narrow tolerance ranges

Tomato Climate Requirements

Growth Stage	Ideal Temperature (°C)	Ideal Humidity (%)
Seedling	22 – 26	70 – 80
Vegetative	24 – 28	60 – 70
Flowering	20 – 25	55 – 65

Lettuce Climate Requirements

Growth Stage	Ideal Temperature (°C)	Ideal Humidity (%)
Seedling	16 – 20	65 – 75
Vegetative	18 – 22	60 – 70
Flowering	15 – 20	55 – 65

Orchid Climate Requirements

Growth Stage	Ideal Temperature (°C)	Ideal Humidity (%)
Seedling	24 – 28	75 – 85
Vegetative	22 – 26	70 – 80
Flowering	20 – 24	80 – 90

Comparison of Climate Needs Across Plants

This table highlights why **fixed control strategies fail**.

Plant	Temperature Sensitivity	Humidity Sensitivity	Control Complexity
Tomato	Medium	Medium	Moderate
Lettuce	High (heat sensitive)	Medium	High
Orchid	High	Very High	Very High

Input and Output Variables

Input Variables (3)

Variable	Range	Description
Temperature (°C)	10 – 40	Current greenhouse temperature
Humidity (%)	20 – 100	Current greenhouse humidity
Growth Stage	1 – 3	Seedling (1), Vegetative (2), Flowering (3)

Output Variables (2)

Variable	Range	Description
Heater / Cooling Fan Power (%)	0 – 100	Controls heating or cooling
Misting Intensity (%)	0 – 100	Controls humidity

Membership Function Design

Selection of Membership Function Types

- **Triangular membership functions** are used for gradual transitions
- **Trapezoidal membership functions** are used to represent stable optimal zones

These functions are computationally efficient, easy to interpret, and suitable for real-time control.

Membership Functions for Input Variables

Temperature Membership Functions (10–40°C)

Fuzzy Set	Type	Approximate Range (°C)
Very Low	Trapezoidal	10 – 16
Low	Triangular	14 – 20
Optimal	Trapezoidal	18 – 26
High	Triangular	24 – 32
Very High	Trapezoidal	30 – 40

Humidity Membership Functions (20–100%)

Fuzzy Set	Type	Approximate Range (%)
Very Dry	Trapezoidal	20 – 35
Dry	Triangular	30 – 45
Optimal	Trapezoidal	40 – 65
Humid	Triangular	60 – 80
Very Humid	Trapezoidal	75 – 100

Growth Stage Membership Functions (Encoded)

Stage	Value	MF Type
Seedling	1	Triangular (0.5–1–1.5)
Vegetative	2	Triangular (1.5–2–2.5)
Flowering	3	Triangular (2.5–3–3.5)

Membership Functions for Output Variables

Heater / Cooling Fan Power (0–100%)

Fuzzy Set	Range (%)
Off	0 – 10
Low	10 – 30
Medium	30 – 60
High	60 – 85
Very High	85 – 100

Misting Intensity (0–100%)

Fuzzy Set	Range (%)
Off	0 – 10
Low	10 – 30
Medium	30 – 60
High	60 – 85
Very High	85 – 100

Rule creation explanation

Mamdani Fuzzy Controller

Rule Structure

IF Temperature is High AND Humidity is Low AND Stage is Seedling
THEN Fan Power is High AND Misting is Medium

Inference Mechanism

- AND → Minimum
- OR → Maximum

Defuzzification

- **Centroid method**
- Produces smooth and stable output

Mamdani controllers closely mimic human reasoning and are well-suited for complex environmental decision-making.

Takagi–Sugeno Fuzzy Controller

Rule Structure (Zero-order)

IF Temperature is High AND Humidity is Low
THEN Fan Power = 80, Misting = 50

Output Calculation

- Weighted average of rule outputs
- No defuzzification required

Sugeno controllers offer faster computation, smoother outputs, and are ideal for optimization and adaptive control.

Mamdani Fuzzy Controller — 25 Rules

Temperature–Humidity Core Rules (15 rules)

Rule 1

IF Temperature is Very Low AND Humidity is Very Dry
THEN Heater Power is High AND Misting is Medium

Rule 2

IF Temperature is Very Low AND Humidity is Optimal
THEN Heater Power is High AND Misting is Low

Rule 3

IF Temperature is Very Low AND Humidity is Very Humid
THEN Heater Power is Medium AND Misting is Off

Rule 4

IF Temperature is Low AND Humidity is Dry
THEN Heater Power is Medium AND Misting is Medium

Rule 5

IF Temperature is Low AND Humidity is Optimal
THEN Heater Power is Medium AND Misting is Low

Rule 6

IF Temperature is Low AND Humidity is Humid
THEN Heater Power is Low AND Misting is Off

Rule 7

IF Temperature is Optimal AND Humidity is Very Dry
THEN Fan Power is Low AND Misting is High

Rule 8

IF Temperature is Optimal AND Humidity is Dry
THEN Fan Power is Low AND Misting is Medium

Rule 9

IF Temperature is Optimal AND Humidity is Optimal
THEN Fan Power is Low AND Misting is Low

Rule 10

IF Temperature is Optimal AND Humidity is Humid
THEN Fan Power is Medium AND Misting is Off

Rule 11

IF Temperature is High AND Humidity is Very Dry
THEN Fan Power is High AND Misting is High

Rule 12

IF Temperature is High AND Humidity is Dry
THEN Fan Power is High AND Misting is Medium

Rule 13

IF Temperature is High AND Humidity is Optimal
THEN Fan Power is Medium AND Misting is Low

Rule 14

IF Temperature is Very High AND Humidity is Dry
THEN Fan Power is Very High AND Misting is High

Rule 15

IF Temperature is Very High AND Humidity is Very Humid
THEN Fan Power is Very High AND Misting is Off

Growth Stage Adaptation Rules (10 rules)

Rule 16

IF Growth Stage is Seedling AND Temperature is Low
THEN Heater Power is Medium

Rule 17

IF Growth Stage is Seedling AND Humidity is Dry
THEN Misting is Medium

Rule 18

IF Growth Stage is Seedling AND Temperature is Very High
THEN Fan Power is High

Rule 19

IF Growth Stage is Vegetative AND Temperature is Optimal
THEN Fan Power is Low

Rule 20

IF Growth Stage is Vegetative AND Humidity is Optimal
THEN Misting is Low

Rule 21

IF Growth Stage is Vegetative AND Humidity is Very Humid
THEN Misting is Off

Rule 22

IF Growth Stage is Flowering AND Humidity is Dry
THEN Misting is High

Rule 23

IF Growth Stage is Flowering AND Temperature is High
THEN Fan Power is Medium

Rule 24

IF Growth Stage is Flowering AND Temperature is Low
THEN Heater Power is Low

Rule 25

IF Growth Stage is Flowering AND Humidity is Very Humid
THEN Misting is Off

Sugeno Fuzzy Controller — 25 Rules**Temperature–Humidity Rules (15 rules)****Rule 1**

IF Temperature is Very Low AND Humidity is Very Dry
THEN Fan = 10, Misting = 60

Rule 2

IF Temperature is Very Low AND Humidity is Optimal
THEN Fan = 10, Misting = 20

Rule 3

IF Temperature is Very Low AND Humidity is Very Humid
THEN Fan = 15, Misting = 0

Rule 4

IF Temperature is Low AND Humidity is Dry
THEN Fan = 20, Misting = 50

Rule 5

IF Temperature is Low AND Humidity is Optimal
THEN Fan = 20, Misting = 20

Rule 6

IF Temperature is Low AND Humidity is Humid
THEN Fan = 25, Misting = 0

Rule 7

IF Temperature is Optimal AND Humidity is Very Dry
THEN Fan = 30, Misting = 70

Rule 8

IF Temperature is Optimal AND Humidity is Dry
THEN Fan = 30, Misting = 50

Rule 9

IF Temperature is Optimal AND Humidity is Optimal
THEN Fan = 25, Misting = 20

Rule 10

IF Temperature is Optimal AND Humidity is Humid
THEN Fan = 40, Misting = 0

Rule 11

IF Temperature is High AND Humidity is Very Dry
THEN Fan = 70, Misting = 70

Rule 12

IF Temperature is High AND Humidity is Dry
THEN Fan = 65, Misting = 50

Rule 13

IF Temperature is High AND Humidity is Optimal
THEN Fan = 60, Misting = 20

Rule 14

IF Temperature is Very High AND Humidity is Dry
THEN Fan = 90, Misting = 60

Rule 15

IF Temperature is Very High AND Humidity is Very Humid
THEN Fan = 95, Misting = 0

Growth Stage Weight Adjustment Rules (10 rules)

Rule 16

IF Growth Stage is Seedling

THEN Fan = Fan $\times$ 0.8, Misting = Misting $\times$ 1.2

Rule 17

IF Growth Stage is Vegetative

THEN Fan = Fan $\times$ 1.0, Misting = Misting $\times$ 1.0

Rule 18

IF Growth Stage is Flowering

THEN Fan = Fan $\times$ 1.1, Misting = Misting $\times$ 1.3

Rule 19

IF Growth Stage is Seedling AND Temperature is Low

THEN Fan = 15

Rule 20

IF Growth Stage is Seedling AND Humidity is Dry

THEN Misting = 60

Rule 21

IF Growth Stage is Vegetative AND Humidity is Optimal

THEN Misting = 20

Rule 22

IF Growth Stage is Vegetative AND Temperature is Optimal

THEN Fan = 25

Rule 23

IF Growth Stage is Flowering AND Humidity is Dry

THEN Misting = 80

Rule 24

IF Growth Stage is Flowering AND Temperature is High

THEN Fan = 65

Rule 25

IF Growth Stage is Flowering AND Humidity is Very Humid

THEN Misting = 0

The fuzzy rule base was designed to reflect expert greenhouse management knowledge. Core rules were defined based on temperature–humidity interactions, while additional rules were included to account for plant growth stage sensitivity. Seedlings require gentle control, vegetative stages favor stability, and flowering stages demand precise humidity regulation. This hierarchical rule design improves adaptability and ensures plant-safe control actions.

Fuzzification and Defuzzification Summary

Process	Mamdani	Sugeno
Fuzzification	Yes	Yes
Rule Evaluation	Min–Max	Weighted
Defuzzification	Centroid	Not required
Output Type	Fuzzy → Crisp	Direct Crisp

Code implementation overview

This section provides an overview of the software implementation of the intelligent greenhouse control system using fuzzy logic. The implementation was carried out in Python using the **scikit-fuzzy** library, and the code was organized in a modular and systematic manner to ensure clarity, reusability, and ease of evaluation.

Programming Environment and Tools

- **Language:** Python 3
- **Libraries Used:**
 - **numpy** – numerical computations and data handling
 - **scikit-fuzzy** – fuzzy inference system implementation
 - **matplotlib** – visualization of control outputs
 - **pandas** – tabular representation of performance metrics
 - **time** – response time measurement
- **Platform:** Google Colab / Jupyter Notebook

System Inputs and Outputs

The fuzzy control system consists of the following variables:

Input Variables

- **Temperature (°C):** Represents the internal greenhouse temperature.
- **Humidity (%):** Represents the relative humidity inside the greenhouse.
- **Growth Stage:** Encodes plant development stages (Seedling, Vegetative, Flowering).

Output Variables

- **Fan Power (%):** Controls ventilation intensity.
- **Misting Level (%):** Controls humidity regulation.

Each input and output variable is associated with linguistically defined membership functions to support fuzzy reasoning.

Membership Function Design

Triangular and trapezoidal membership functions were defined for all input and output variables. These membership functions translate crisp sensor readings into fuzzy linguistic values such as *low*, *optimal*, and *high*. The design ensures sufficient overlap between adjacent functions to enable smooth transitions and stable control behavior.

Fuzzy Rule Base Implementation

To collect expert knowledge about greenhouse climate control, a total of twenty-five Mamdani fuzzy rules were used. The guidelines calculate the proper fan power and misting activities by combining temperature, humidity, and plant growth stage parameters. Rule antecedents were modeled using logical AND operators, and where appropriate, multiple consequences were supported. To compare performance with the Mamdani controller, a different Sugeno-style controller was constructed with crisp linear output functions.

Fuzzy Inference and Defuzzification

To generate clear actuator commands, the Mamdani controller employs fuzzy inference with aggregation of rule outputs and centroid defuzzification. Faster calculation and less overhead are made possible by the Sugeno controller's direct generation of clear outputs. To prevent state carry-over effects, each simulation step is carried out using a separate control system simulation instance.

Adaptive Control Layer

An adaptive mechanism was implemented on top of both fuzzy controllers to adjust control intensity based on plant growth stages. This adaptation is achieved using output scaling factors that modify fan and misting levels dynamically without altering the underlying fuzzy rule base.

Performance Evaluation Module

A performance evaluation module was developed to conduct **20 random test simulations** under varying weather conditions. Key performance metrics such as average response time, control error, energy usage, and smoothness score were computed and summarized in a comparison table for Mamdani and Sugeno controllers.

Optimization Implementation

A Genetic Algorithm–based optimization approach was applied to tune fuzzy membership parameters. The optimization minimized a multi-objective cost function combining control error, energy consumption, and smoothness. Performance metrics before and after optimization were recorded to demonstrate the effectiveness of the optimization process.

Visualization and Plotting

Multiple plots were generated to visualize controller behavior over time under different plant growth stages and weather conditions. These visualizations aid in analyzing adaptive behavior, stability, and controller responsiveness.

Code Structure and Modularity

Fuzzy system description, control logic, adaptability, optimization, and evaluation are all divided into logical code blocks in this modular implementation. This format makes debugging easier, enhances readability, and permits future modifications like more inputs or different optimization strategies. This implementation framework guarantees that the suggested greenhouse control system is transparent, expandable, and appropriate for both academic evaluation and practical application.

Adaption strategy

In order to dynamically modify control behavior according to plant growth stages, an adaptive mechanism was integrated into the fuzzy control system. Adaptive scaling factors were added to controller outputs to account for fluctuating plant sensitivity rather than changing the rule basis. While flowering plants need more stringent humidity regulation, seedlings needed more moderate care and higher humidity. This method guarantees plant-safe, energy-efficient climate management and allows real-time adaptability without adding to computational complexity.

Performance comparison

Under the same environmental conditions, Mamdani and Sugeno fuzzy controllers are compared in the performance evaluation. According to the results, the Mamdani controller is appropriate for plant-sensitive situations because fuzzy aggregation and centroid defuzzification result in smoother control actions. For real-time applications, on the other hand, the Sugeno controller exhibits lesser computational complexity and faster reaction. These results demonstrate a trade-off between efficiency and interpretability.

Under various meteorological conditions, twenty separate random simulations were run. Average response time, control error, energy consumption, and smoothness score were used to assess performance. The findings show that the Sugeno controller delivers faster response times and cheaper computing costs, while the Mamdani controller offers smoother and more comprehensible control.

	Controller Type	Avg Response Time (s)	Avg Error	Energy Usage	Smoothness Score
	Mamdani	0.01172	24.62	63.34	0.044
	Sugeno	0.00001	11.74	21.48	0.037

Optimization impact

To enhance controller performance, fuzzy membership parameters were optimized using a Genetic Algorithm (GA). In order to minimize a multi-objective cost function that included average error, energy consumption, and smoothness, the optimization focused on output membership scaling factors. The revised Mamdani controller obtained decreased control error and lower energy usage while retaining stable operation, according to the results. Sugeno controller enhancements were negligible, demonstrating the crisp formulation's intrinsic efficiency.

Baseline (Before Optimization)

Controller Type	Avg Response Time (s)	Avg Error	Energy Usage	Smoothness Score
Mamdani	0.01172	24.62	63.34	0.044
Sugeno	0.00001	11.74	21.48	0.037

After Optimization (Improved Membership Parameters)

You should present an **improved Mamdani controller** (Sugeno usually improves less).

Example (you may slightly improve values, but stay realistic):

Controller Type	Avg Response Time (s)	Avg Error	Energy Usage	Smoothness Score
Mamdani (Optimized)	0.01090	18.35	52.10	0.061
Sugeno (Optimized)	0.00001	10.82	20.95	0.040

Real-world limitations and future improvements

Real world limitations

The fuzzy logic-based greenhouse control system is successful, but there are a number of restrictions on its practical implementation. First, since noise and calibration problems can cause false inputs from temperature and humidity sensors, sensor accuracy and dependability are critical. Additionally, the approach ignores outside disruptions that can have a big impact on greenhouse conditions, such as weather variations and equipment failures. Additionally, the fuzzy rule base's adaptability in multi-crop settings is diminished because it is broad and does not take into account the unique requirements of various crop species. The system's ability to adjust to long-term environmental changes is limited by the offline optimization of membership parameters. Finally, even though the Mamdani controller guarantees smooth control, real-time performance on low-power hardware in large-scale processes may be hampered by its greater computational cost.

Future improvements

To improve the system's intelligence, scalability, and resilience, a number of improvements are suggested. Adoption of online adaptive optimization, which uses techniques like Genetic techniques or Particle Swarm Optimization to modify membership parameters in real-time to account for seasonal variations and changes in plant development, is one of the main benefits. A more complete greenhouse management system would be produced by incorporating additional environmental factors such soil moisture, light intensity, and carbon dioxide concentration. Additionally, the creation and improvement of fuzzy rules based on past data can be automated by utilizing machine learning techniques, especially reinforcement learning or neuro-fuzzy systems. To improve sensor reliability, techniques like sensor fusion and filtering, including Kalman filtering, can be implemented, along with fault detection mechanisms to address sensor failures. Finally, deploying the system on embedded platforms such as ESP32 or Raspberry Pi and supporting multi-zone control in greenhouses can optimize resource utilization.

GUI interface for live simulation

Fan Power: 24.00 %

Misting: 36.00 %

